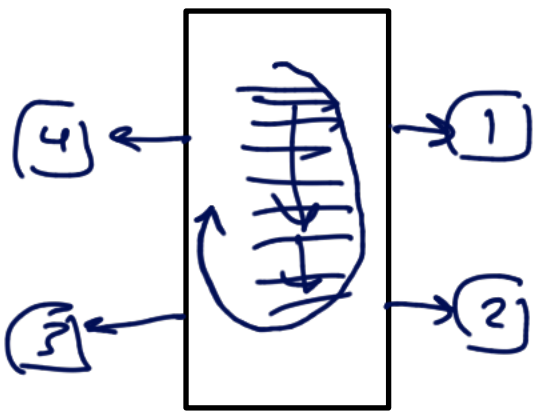
ch 11
interrupt

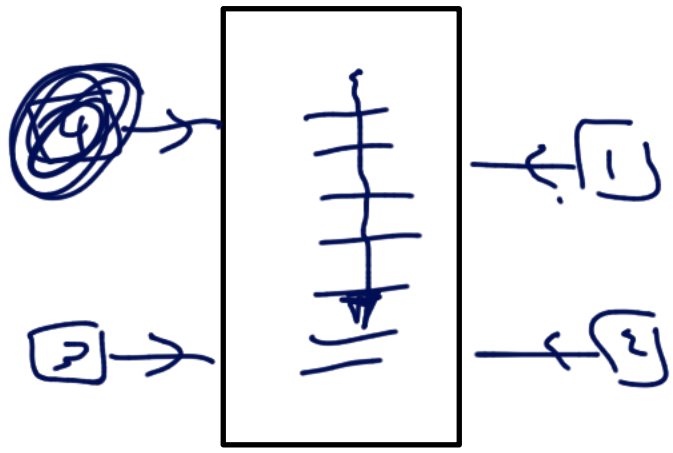
- 8051 can services many devices in two ways

① polling



- It wastes the 8051 (CPU) time
- There is no priority (round robin fashion)

② Interrupt



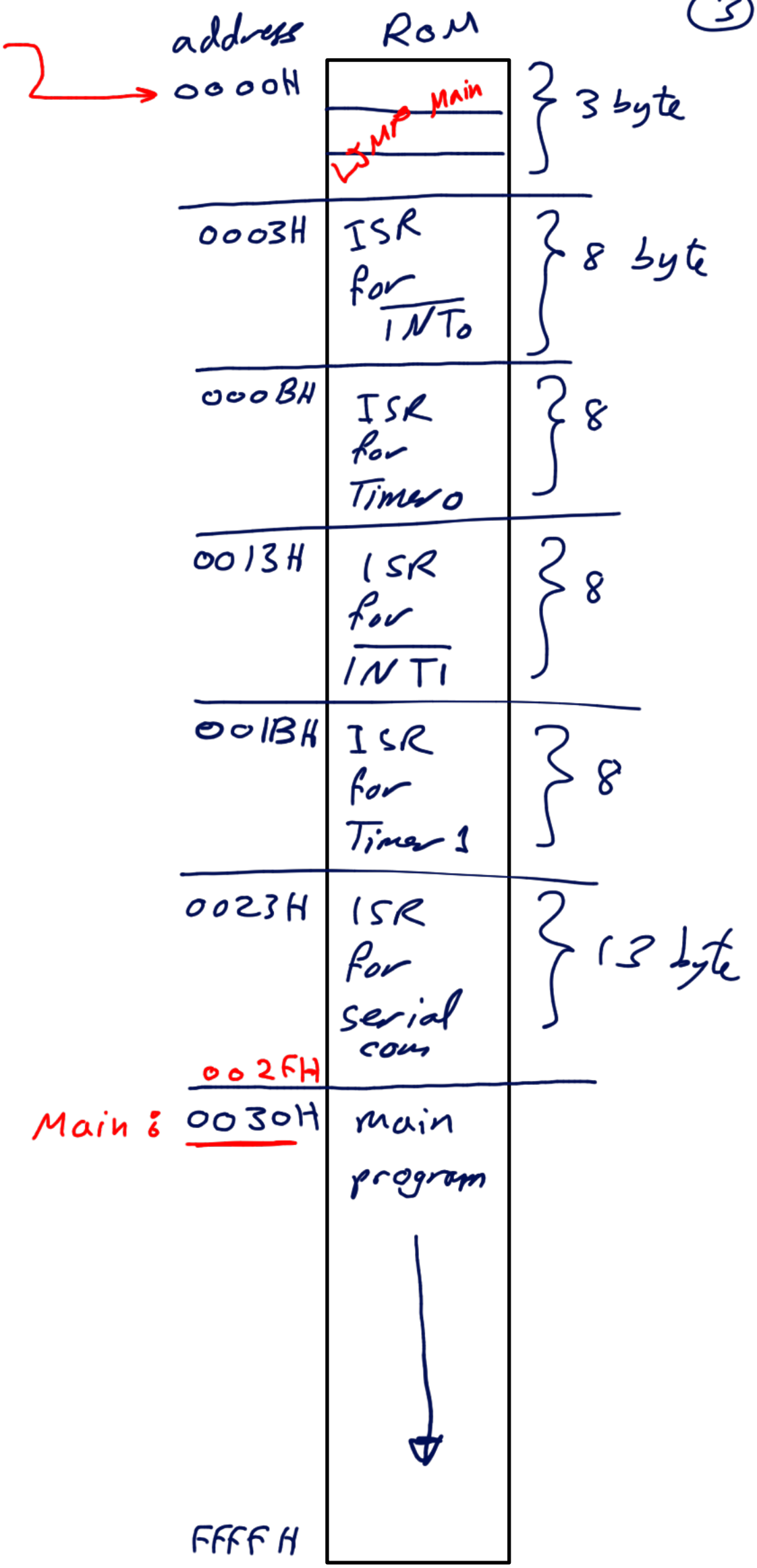
- It save the 8051 time
- There is a priority (programmable)

- ISR: Interrupt service routine (subroutine)
- 8051 has 6 types of interrupts (only 5 are programmable)
- For each interrupt, there must be an ISR.

- For every ISR there is a fixed locations in memory. (ROM).
- The group of memory locations that hold the addresses of ISRs is called Interrupt vector table.

Address	interrupt
0000H	Reset Interrupt
0003H	External hardware 0 interrupt (INT0) P3.2 (EX0)
000BH	Timer0 interrupt (TF0)
0013H	External hardware 1 interrupt (INT1) P3.3 (EX1)
001BH	Timer1 interrupt (TF1)
0023H	serial interrupt

by default PC = 0000H



```
ORG 0000H
LJMP Main

ORG 000BH
MOV R2,#15
MOV A,R2
MOV B,#10
RETI

ORG 0030H
Main: MOV A,R3
      INC A
      :
      MOV R7,#16H
      MOV B,R7
      :
      SJMP $
```

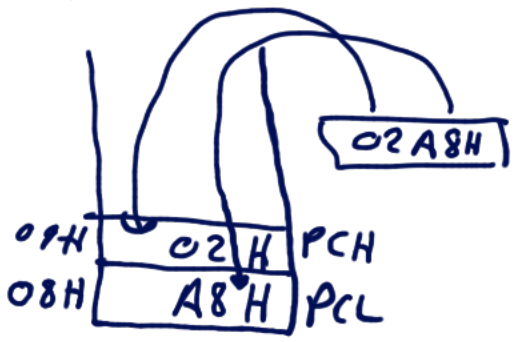
ISR

main program

R7=16H

① finish
R7 = 16H

② stack



interrupt

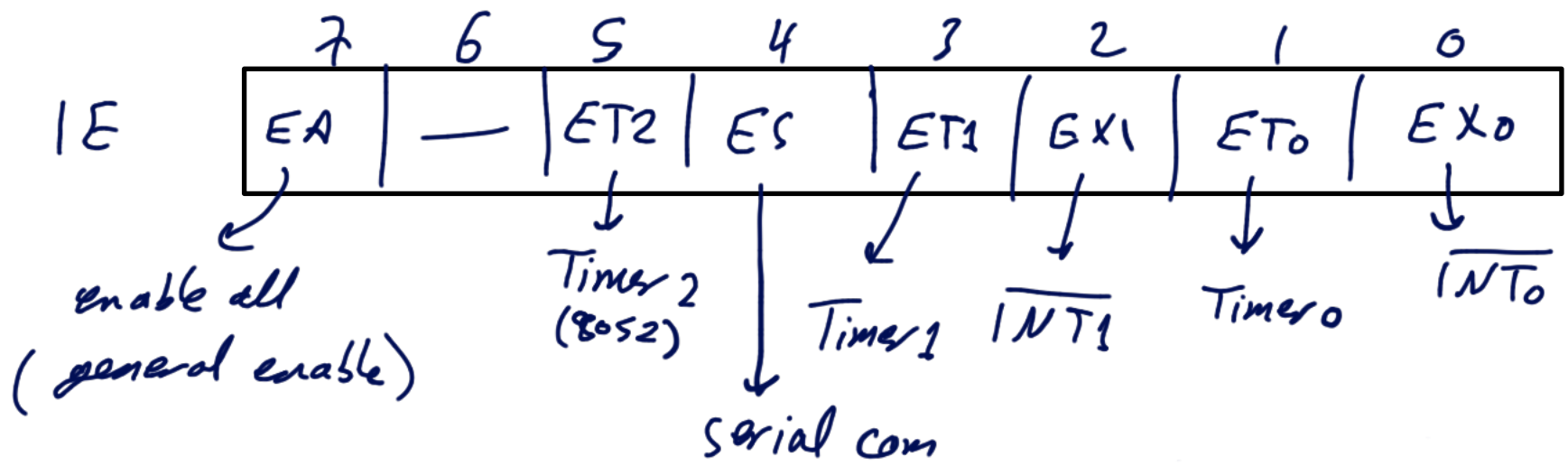
02A8H → PC

③ → Interrupt vector table
000BH

④ → execute ISR

⑤ → RETI
pop two bytes from stack into PC. To return the control to main program

- By default all interrupts are disabled.
- Programmer can enable / disable any interrupt using IE register (Interrupt Enable) → is bit addressable



Enable timer 0 interrupt and serial com interrupt

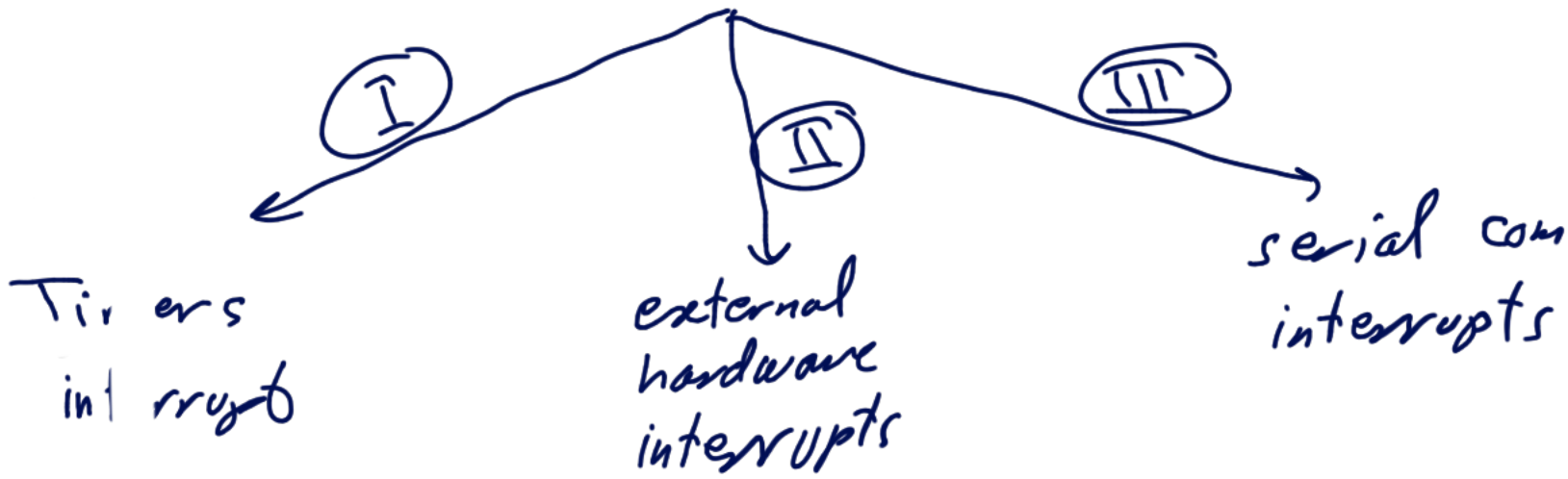
```
MOV  IE,#10010010B
MOV  IE,#92H
```

```
SETB IE.1
SETB IE.4
SETB IE.7
```

Disable serial com interrupt

```
MOV  IE,#10000010B
CLR  IE.4
```

Interrupt programs

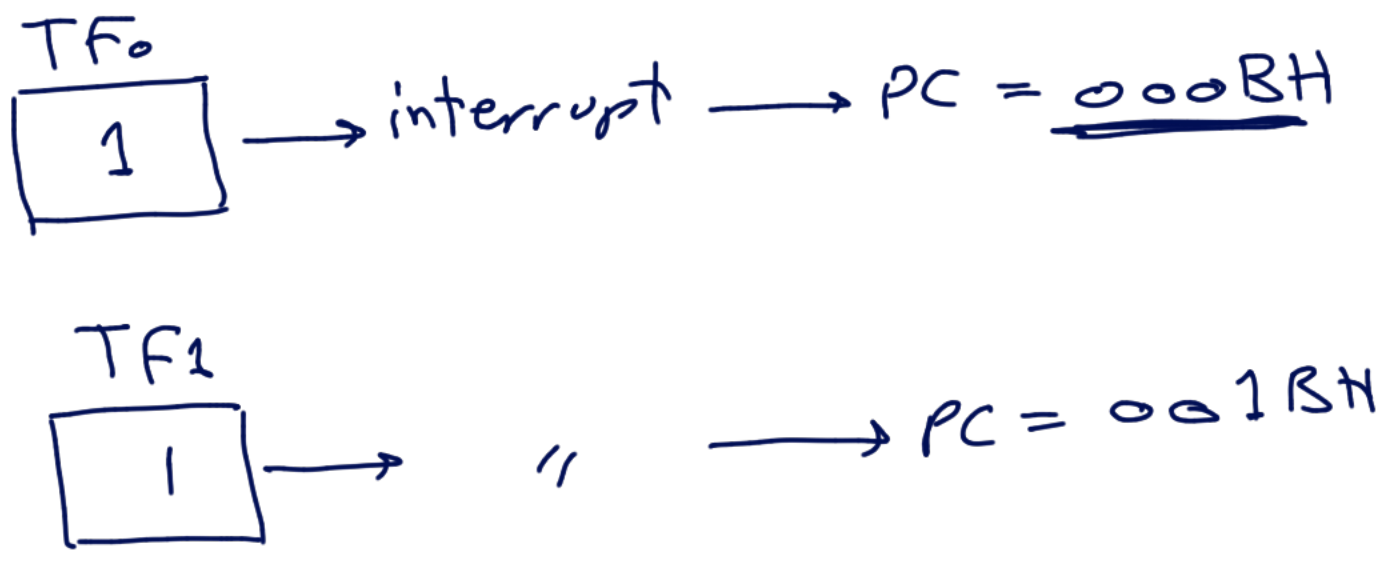


(I) Timer Interrupts:

* polling timer: 8051 will continuously monitor TFi

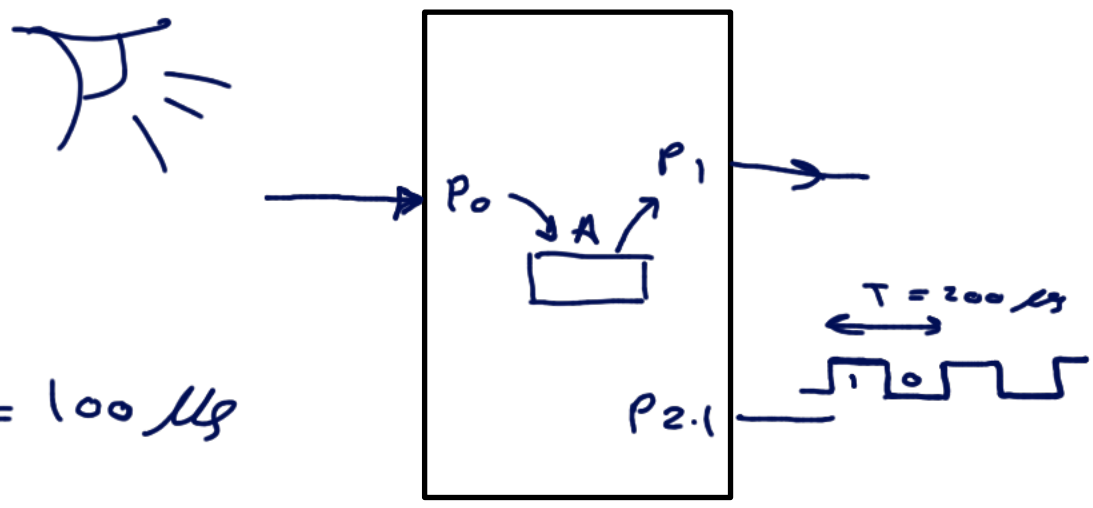
```
HERE: JNB  TFi, HERE
          or 1
```

* Interrupt:



Write a program that continuously gets 8 bit data from P_0 and send it to P_1 while simultaneously creating a square wave of 200 μs period on P2.1. Use Timer 0. Assume XTAL = 12 MHz.

$$MC = \frac{12}{12\,MHz} = 1\,\mu s$$



$$Delay = \frac{T}{2} = \frac{200\,\mu s}{2} = 100\,\mu s$$

$$\frac{Delay}{MC} = \frac{100\,\mu s}{1\,\mu s} = 100 < 256 \longrightarrow \text{mode 2 Timer 0}$$

@ ⟶ MOV TH0, #−100

```
        ORG   0000H
        LJMP  Main

        ORG   000BH
        CPL   P2.1
        RETI

        ORG   0030H
Main:   MOV   P0, #0FFH  ; input
        MOV   TMOD, #0000 0010B
        MOV   TH0, #-100
        MOV   IE, #10000010B
        SETB  TR0
Back:   MOV   A, P0
        MOV   P1, A
        SJMP  Back
```

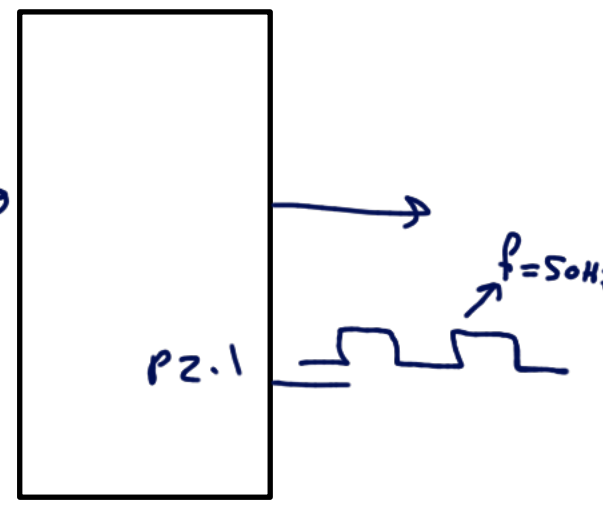
TF0 = 1

- Write a program to generate square wave of 50 HZ on P2.1 using interrupt. use timer 1.
Assume crystal frequency = 24 MHz.

$$MC = \frac{12}{24\,MHz} = 0.5\ \mu s$$

$$T = \frac{1}{f} = \frac{1}{50\,Hz} = 0.02\ sec$$

$$Delay = \frac{T}{2} = \frac{0.02\ sec}{2} = 0.01\ sec$$



$$\frac{Delay}{MC} = \frac{0.01\ sec}{0.5\ \mu s} = \underline{20\,000} > 256 \rightarrow \text{mode } 1$$

$65536 - 20000 = 45536$

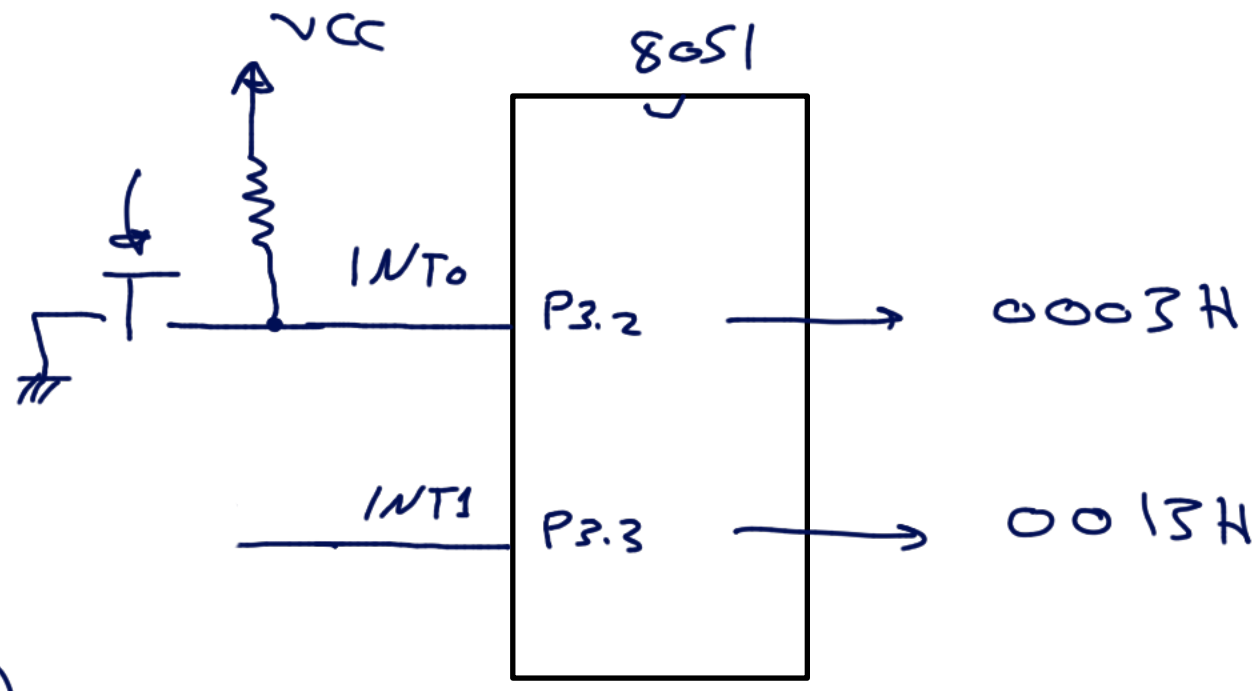
$\rightarrow = B1E0H$

TH1 TL1

```
        ORG  0000H
        LJMP Main

        ORG  001BH
        LJMP SQR

        ORG  0030H
Main :  MOV  TMOD, #0001 0000B      → 10H
        MOV  TL1, #0E0H
        MOV  TH1, #0B1H             → 88H
        MOV  IE, #10001000B

        SETB TR1
HERE:   SJMP HERE

SQR:    CLR  TR1
        CPL  P2.1
        MOV  TL1, #0E0H
        MOV  TH1, #0B1H
        SETB TR1
        RETI
```

(II) External Interrupt

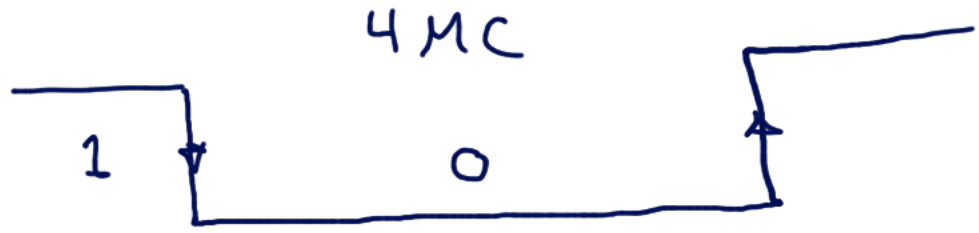
- 8051 has two external interrupts:



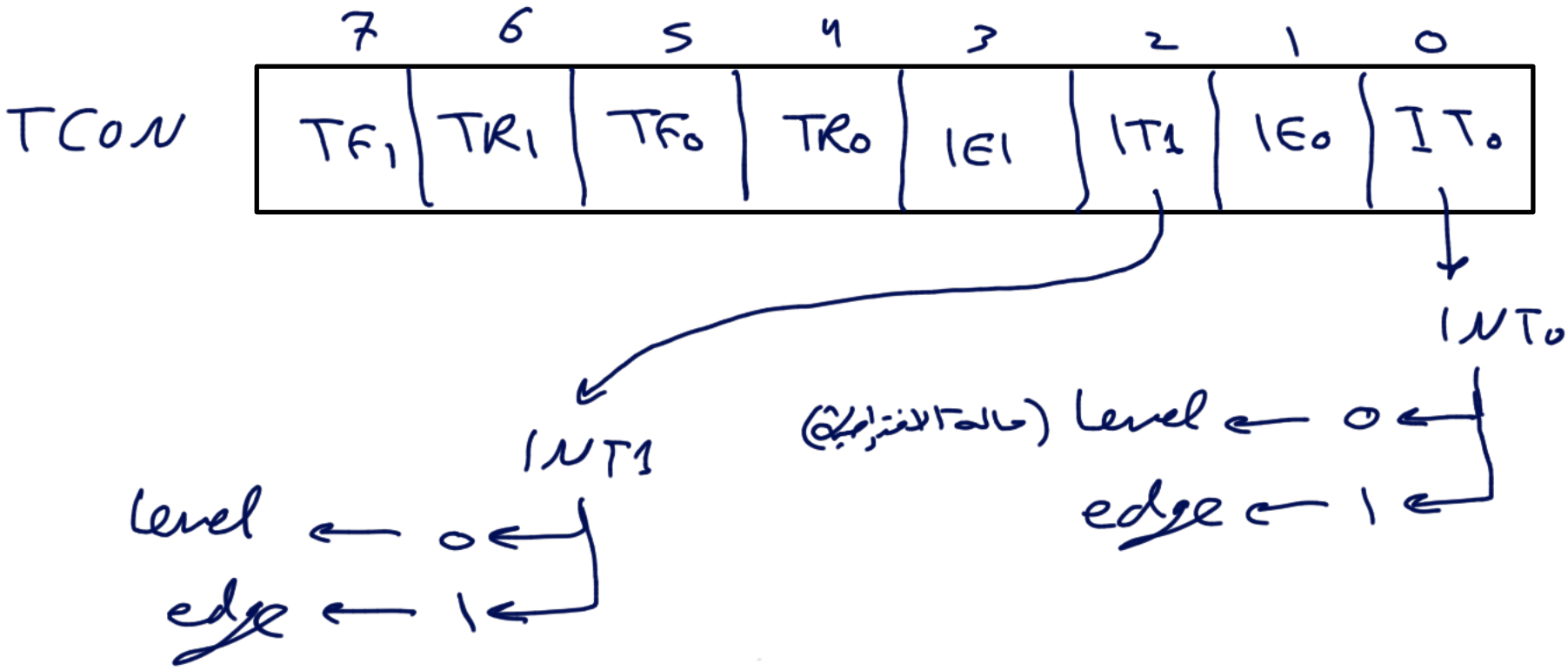
P3.2 must be 1 } no interrupt
P3.3 " " 1

There are two ways for activating external interrupt

① Level triggered (by default)



② Edge triggered
(falling edge)

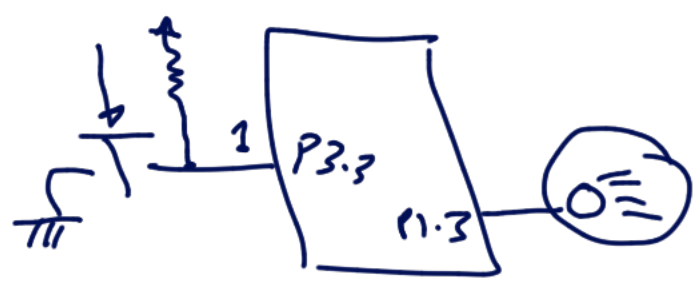


activate external interrupt 0 to be edge

SETB TCON.0

INT1 pin is connected to switch that is normally high. whenever it goes low (0) it should turn on LED on (P1.3). when it is turned on it should stay on for a fraction of seconds. As long as the switch is pressed low the LED should stay on.

(level)



```
        ORG  0000H
        LJMP Main

        ORG  0013H
        LJMP LEDON

        ORG  0030H
Main :  CLR  P1.3  ; LED is off
        MOV  IE,#10000100B
        SJMP $

LEDON : SETB P1.3
        MOV  R2,#250
HERE :  DJNZ R2,HERE
        CLR  P1.3
        RETI
```

SETB TCON.2 →

← LEDON

} delay

← CLR P1.3

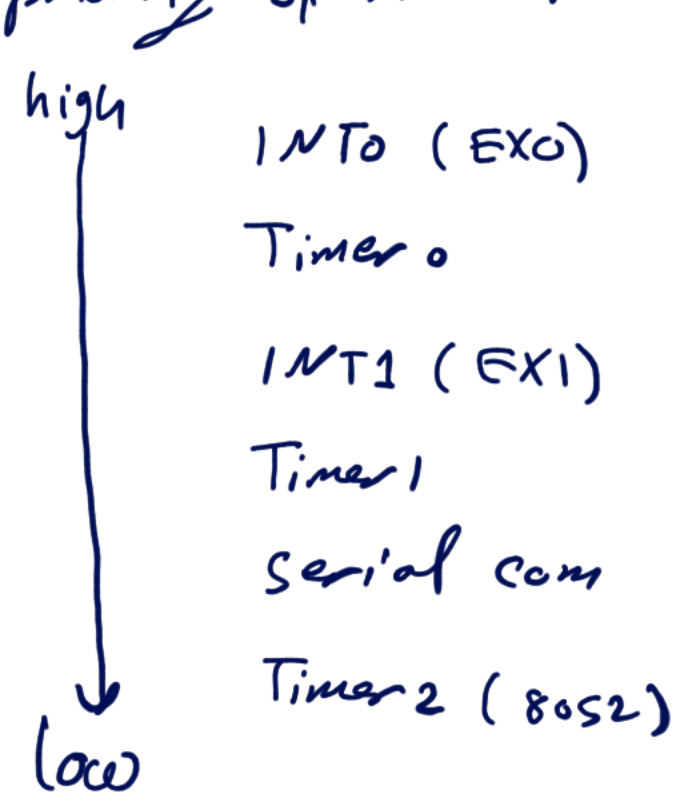
compare between RET and RETI ?

RET: return to main program only.

RETI: return to main program in addition to clear TF0 and TF1 in timers interrupts and clear TCON.1 (ET0) and TCON.3 (ET1) in edge triggered external interrupts.

Interrupt priority

Default priority of interrupts



- programmer can change priority using IP

IP : Interrupt priority (8 bit)

IP

7	6	5	4	3	2	1	0
—	—	PT_2	PS	PT_1	PX1	PT_0	PX0

by default IP = 00H

---

configure Timer2 interrupt to be the highest priority

MOV IP, #00001000B

Timer1, INT0, Timer0, INT1, Serial, Timer2

MOV IP, #00001010B

Timer0, Time1, INT0, INT1, Serial, Timer2

MOV IP, #00111111B

↓

by default priority